

Incidental Learning Through Conversation With Professionals

A Proposal to Connect the School Curriculum to Engineering, Health, and Economics

Miebale Nigus

Summary

School teaches many chapters without showing what they are for. Students study without purpose or plan, and choose careers, especially white-collar ones, knowing little more than their names. This proposal adds a second method of learning alongside school: small groups of students discuss one real concern in engineering, health, or economics with a professional who works on it. New words and ideas are learned incidentally as questions arise, but only within that concern, and are then linked to the curriculum and to the professional's actual work. Delivery is in person, through an online mobile app in cities, and through an offline app in rural areas.

1. The Problem

1. **Learning without relevance.** In Ethiopia and many other countries, students learn courses and chapters with no visible connection to their lives. I learned mathematics without knowing it leads to computer science, and C3 and C4 photosynthesis without knowing it explains which crops survive a dry season.
2. **Learning without purpose or plan.** Students who cannot see where a subject leads study to pass exams rather than toward a goal of their own, with no plan for what to focus on or why.
3. **Careers known only by name.** Students rarely know what most professions involve, especially white-collar ones such as accountant, pharmacist, economist, or software engineer. Unlike farming or shopkeeping, this work happens in offices and on screens that students never see. They choose university departments by name, and many discover years later that the name did not match the work.

2. The Approach

Sessions cover three domains, **engineering, health, and economics**, which draw on the school subjects of mathematics, physics, chemistry, biology, geography, and economics, and lead to a large share of professional careers. Groups of eight to twelve secondary school students meet for about an hour every two weeks, alongside school, rotating through the three domains. Each session has four steps:

1. **Concern:** one real question from one domain.
2. **Conversation:** a professional who works on that question discusses it in their normal working language, without simplifying. Students ask whenever a word or idea is unfamiliar, and the professional explains.
3. **Curriculum link:** a facilitator, usually a teacher, connects what came up to the school subjects and chapters where it appears.
4. **Career view:** the professional describes the actual work: daily tasks, tools, decisions, and the path into the job.

Domain	Concern	Professional	Curriculum link
Engineering	Why do some buildings crack after the rainy season?	Civil engineer	Physics: force, pressure. Geography: soil types.
Health	How does a doctor find the cause of a fever?	Doctor	Biology: blood, microorganisms, the immune system.
Economics	Why did food prices rise this year?	Economist	Mathematics: percentages, graphs. Economics: supply and demand.

In the health session, learning happens like this:

Doctor: "Malaria, typhoid, and flu can all start with a fever, so I test the blood before I diagnose."

Student: "What does diagnose mean?"

Doctor: "To identify the illness from the symptoms and the test results."

The word was learned because the concern required it, not because it was on a list. The facilitator then links the discussion to the biology chapters on blood and microorganisms, and the doctor describes a working day: examining patients, ordering tests, reading results, and deciding treatment.

3. Where Incidental Learning Fits

Incidental learning is a second method alongside school, not a replacement for it. It operates only within the session's concern, never as open-ended general education. Three rules keep it that way:

- **Stay on the concern.** Questions that lead elsewhere are noted for a later session where they belong, not followed.
- **Converse, do not lecture.** Professionals explain what students ask, not what a syllabus would cover.
- **No tests or grades.** Not knowing is a reason to ask, not a reason to be quiet.

The facilitator's main task is to prevent the two usual drifts: into a lecture, and away from the concern.

4. Worldwide Delivery

The same session design runs in person and digitally, in urban and rural settings:

	Urban	Rural
In person	Groups meet at schools or at workplaces such as clinics, banks, and engineering offices.	Groups meet at schools or local clinics with professionals working in the area: health workers, agricultural experts, cooperative accountants, and engineers on local projects. Where none is available, a teacher plays a recorded session from the offline app and facilitates.
Digital	Online app: live sessions with professionals, a library of recorded sessions, and student questions answered by professionals.	Offline app: session packs downloaded once at school or shared between phones, and used on older phones without mobile data. Questions are saved offline and answered when the phone next connects.

- **Shared worldwide:** the session design, the library of concerns, and recorded conversations with professionals.
- **Adapted locally:** language, examples, professionals, and curriculum links for each country's subjects and grades.

Schools host groups, professional associations supply professionals, and teachers map sessions to the national curriculum. Online sessions can draw on professionals from any country, including the diaspora. Offline delivery is already practical: Ahazawi, an offline-first learning app I helped build, serves more than 6,500 learners on older phones without reliable internet.

5. Expected Outcomes

Four outcomes are measured at the start and end of the year. The first three answer the three problems; the last measures incidental learning itself:

- **Relevance:** students can explain where school topics are used in real work.
- **Direction:** students can state what they want to study, why, and which subjects lead there.
- **Career knowledge:** students can describe what specific professions do day to day, not only name them.
- **Vocabulary:** students understand and use the terms that arose in sessions.

6. Pilot

Run the program for one school year in two Ethiopian schools: one urban school using in-person sessions and the online app, and one rural school using in-person sessions and the offline app. Record which concerns hold attention, how often students ask questions, whether sessions stay on their concern, and how in-person and app sessions compare. The results will show what to change before expanding to more schools and countries. I intend to develop this design during graduate study and to build the app for urban and rural students alike, so that learning starts from what a topic is **for**.